



DAY 5 – Functions in Python

Functions are one of the most important Python concepts because you'll use them throughout AI/ML projects to organize code, reuse logic, and build machine learning pipelines.



What is a Function?

A function is a reusable block of code that performs a specific task.

Instead of writing the same code multiple times, you write it once and call it whenever needed.

Example:



```
def greet(): print("Hello, World!") greet()
```

Output:



Hello, World!

Function Syntax



```
def function_name(): # code  
function_name()
```

Function with Parameters

Parameters allow us to pass information into a function.



```
def greet(name): print("Hello", name)
greet("Utkarsh")
```

Output



Hello Utkarsh

Multiple Parameters



```
def add(a, b): print(a + b) add(5, 7)
```

Output

12



Return Statement

return sends a value back from the function.

```
def square(num): return num * num  
result = square(6) print(result)
```



Output

36



Difference Between print() and return

Using print():

```
def add(a, b): print(a + b) add(2, 3)
```



Using return:

```
def add(a, b): return a + b answer =  
add(2, 3) print(answer)
```



return is preferred because the returned value can be reused.

Default Parameters



```
def greet(name="Student"):
    print("Hello", name)
greet()
greet("Utkarsh")
```

Output



Hello Student Hello Utkarsh

Keyword Arguments



```
def student(name, age): print(name,  
age) student(age=20, name="Utkarsh")
```

Output



Utkarsh 20

Variable Scope

Local variables exist only inside a function.



```
def demo(): x = 10 print(x) demo()
```

Global variables can be accessed outside functions.

```
x = 50 def show(): print(x) show()
```



Output

50



Lambda Functions

A lambda function is a short anonymous function.



```
square = lambda x: x * x  
print(square(8))
```

Output



64

Function Calling Another Function



```
def greet(): print("Hello")  
def welcome(): greet() print("Welcome")  
welcome()
```

Output



Hello Welcome

Recursion

A recursive function calls itself.



```
def countdown(n): if n == 0: return  
print(n) countdown(n - 1)  
countdown(5)
```

Output



5 4 3 2 1

Mini Project 1 – Calculator



```
def add(a, b): return a + b
def subtract(a, b): return a - b
print(add(10, 5))
print(subtract(10, 5))
```

Mini Project 2 – Even or Odd



```
def even_or_odd(num): if num % 2 ==  
0: return "Even" return "Odd"  
print(even_or_odd(9))
```

Mini Project 3 – Student Grade



```
def grade(marks): if marks >= 90:  
return "A" elif marks >= 75: return "B"  
elif marks >= 50: return "C" else: return  
"Fail" print(grade(84))
```

Interview Questions

Q1. What is a function?

A reusable block of code that performs a specific task.

Q2. Why do we use functions?

- Reduce code duplication

- Improve readability

- Make programs modular

- Simplify debugging

Q3. What is the difference between parameters and arguments?

Parameters are variables defined in the function.

Arguments are the actual values passed when calling the function.

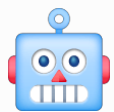
Q4. What is the difference between print() and return?

print() displays output.

return sends a value back to the caller for further use.

Q5. What is recursion?

A technique where a function calls itself until a stopping condition (base case) is met.



AI/ML Connection

Functions are everywhere in AI and machine learning:



Preprocessing datasets



Cleaning data



Building training pipelines



Calculating evaluation metrics



Reusing model training and prediction code



Organizing large AI projects into reusable modules



Practice Challenge

Build a Student Result System using functions:

Create a function to input marks.

Create a function to calculate the total and average.

Create a function to assign a grade.

Display the complete result using function calls.

This will strengthen your understanding of functions before moving on to file handling and exception handling in the next day's roadmap.

